Quadratic-size discretization of bounded continuous frames with stable real phase retrieval

Autonomous solution packet for arXiv:2109.14454

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Status: candidate partial result, likely valid. The original problem asks for $O(N)$ samples. We prove an unconditional $O_{C,\beta}(N^2)$ theorem over the real field and recover $O(N)$ under a uniform joint small-ball hypothesis. The complex case is not claimed.

1 Problem and result

Let H be a real N-dimensional Hilbert space and let $(x_t)_{t\in\Omega}$ be a continuous Parseval frame over a probability space:

$$\int_\Omega |\langle x, x_t\rangle|^2 \, d\mu(t) = \|x\|^2.$$

It does C-stable phase retrieval if

$$\min\{\|x-y\|, \|x+y\|\} \le C \left(\int_\Omega \big(|\langle x, x_t\rangle| - |\langle y, x_t\rangle|\big)^2 \, d\mu(t) \right)^{1/2}. \tag{SPR}$$

Problem 4.5 of Freeman–Ghoreishi asks whether, under $\|x_t\| \le \beta\sqrt{N}$, one can always retain dimension-free stability using $M = O(N)$ equally weighted samples. The source writes C in the conclusion although it introduces a new constant κ; Question 7.2 of Freeman–Oikhberg–Pineau–Taylor (arXiv:2210.05114) confirms that κ is intended.

Theorem 1 (quadratic sampling)**.** *There is an absolute constant K with the following property. Suppose $N \ge 2$, (x_t) is a continuous Parseval frame of a real N-dimensional Hilbert space, it satisfies* (SPR)*, and $\|x_t\| \le \beta\sqrt{N}$ for every t. Then there are*

$$M \le K\beta^2 C^2 N^2 \log(2 + C^2)$$

sampling points $t_1, \dots, t_M$ (repetitions allowed) such that $(M^{-1/2} x_{t_j})_{j=1}^M$ is a frame and does $2C$-stable phase retrieval. Its lower frame bound is at least $1/(4C^2)$ and its upper frame bound is at most 2.

For fixed C, β, this is $O(N^2)$. The immediate pointwise boundedness and net argument gives $O(N^2 \log N)$; controlling the empirical frame operator first is what removes $\log N$.

2 The min-of-two-quadratics reduction

For unit vectors $u, v \in H$, define

$$q_{u,v}(t) = \min\{|\langle u, x_t\rangle|^2, |\langle v, x_t\rangle|^2\}.$$

For real scalars a, b,

$$\left(\frac{|a+b|-|b-a|}{2}\right)^2 = \min\{a^2, b^2\}. \tag{1}$$

Apply (SPR) to $x = (u+v)/2$ and $y = (v-u)/2$. Since $x - y = u$ and $x + y = v$, (1) gives

$$\int_\Omega q_{u,v}(t)\, d\mu(t) \geq c, \qquad c = C^{-2}, \tag{2}$$

for every unit pair. Also

$$0 \leq q_{u,v}(t) \leq B, \qquad B = \beta^2 N. \tag{3}$$

Notice that Parseval normalization implies $C \geq 1$ and $\beta \geq 1$.

3 Proof of Theorem 1

Proof. Choose $t_1, \ldots, t_M$ independently according to μ and put

$$S_M = \frac{1}{M}\sum_{j=1}^{M} x_{t_j} x_{t_j}^*.$$

The matrices in the sum are positive, bounded above by BI, and have mean I. Matrix Chernoff therefore gives an absolute $a > 0$ such that

$$\mathbb{P}\{\|S_M\| > 2\} \leq N \exp(-aM/B). \tag{4}$$

Let $\delta = c/64$ and take a δ-net $\mathcal{N}$ of the unit sphere with

$$|\mathcal{N}| \leq (1 + 2/\delta)^N.$$

For fixed $u_0, v_0 \in \mathcal{N}$, write $Z_j = q_{u_0,v_0}(t_j)$. By (2)–(3), $0 \leq Z_j \leq B$ and $\mathbb{E}Z_j \geq c$. The multiplicative scalar Bernstein inequality yields

$$\mathbb{P}\left\{\frac{1}{M}\sum_{j=1}^{M} Z_j < \frac{c}{2}\right\} \leq \exp(-Mc/(8B)). \tag{5}$$

Taking a union bound over $\mathcal{N}^2$, the probability that (5) fails for some net pair is at most

$$(1 + 2/\delta)^{2N} \exp(-Mc/(8B)). \tag{6}$$

For a sufficiently large absolute K, the asserted choice of M makes the sum of (4) and (6) strictly less than one. Fix a realization for which

$$S_M \leq 2I, \qquad \frac{1}{M}\sum_{j=1}^{M} q_{u_0,v_0}(t_j) \geq c/2 \quad (u_0, v_0 \in \mathcal{N}). \tag{7}$$

We extend (7) from the net without paying a dimension-dependent Lipschitz constant. If u, u_0 are unit vectors, Cauchy–Schwarz and $S_M \leq 2I$ give

$$\begin{aligned}\frac{1}{M}\sum_j \left||\langle u, x_{t_j}\rangle|^2 - |\langle u_0, x_{t_j}\rangle|^2\right| &\leq \left(\frac{1}{M}\sum_j |\langle u + u_0, x_{t_j}\rangle|^2\right)^{1/2} \left(\frac{1}{M}\sum_j |\langle u - u_0, x_{t_j}\rangle|^2\right)^{1/2} \\ &\leq 2\|u + u_0\|\,\|u - u_0\| \leq 4\|u - u_0\|.\end{aligned} \tag{8}$$

The elementary inequality $|\min(a,b)-\min(a',b')|\le|a-a'|+|b-b'|$, followed by (8), shows that replacing both members of a unit pair by nearest net points changes the empirical average of q by at most $8\delta=c/8$. Hence

$$\frac{1}{M}\sum_{j=1}^{M}q_{u,v}(t_j)\ge c/4\quad\text{for every unit } u,v. \tag{9}$$

Now take arbitrary $x,y\in H$ and set $u=x-y$, $v=x+y$, and $s=\min\{\|u\|,\|v\|\}$. If $s>0$, normalize u and v separately. Since both scaling factors are at least s, (1) and (9) imply

$$\frac{1}{M}\sum_{j=1}^{M}\big(|\langle x,x_{t_j}\rangle|-|\langle y,x_{t_j}\rangle|\big)^2\ge\frac{c}{4}s^2.$$

Thus the sampled frame does $2C$-stable phase retrieval. Taking $y=0$ gives its lower frame bound $c/4$, while $S_M\le 2I$ gives upper bound 2. □

4 When the desired linear bound follows

The quadratic proof loses a factor N because (2) permits the expectation of a particular $q_{u,v}$ to be carried by an event of probability $O(1/N)$ and height $O(N)$. If this behavior is excluded, the original linear conclusion follows.

Proposition 1 (joint small-ball criterion)**.** *Assume that there are dimension-free $\alpha,p>0$ such that*

$$\mu\{t:q_{u,v}(t)\ge\alpha\}\ge p\quad\textit{for every unit } u,v. \tag{10}$$

Then there are $M\le K_{\alpha,p}N$ samples for which $(M^{-1/2}x_{t_j})$ is a frame and does $\sqrt{2/(\alpha p)}$-stable phase retrieval.

Proof. The ranges

$$\{t:|\langle u,x_t\rangle|\ge\sqrt{\alpha},\ |\langle v,x_t\rangle|\ge\sqrt{\alpha}\}$$

form a semi-algebraic range class of VC dimension at most K_0N. The VC uniform convergence theorem, applied with additive error $p/2$, gives $M\le K_{\alpha,p}N$ points for which every range in (10) has empirical frequency at least $p/2$. Therefore every unit pair satisfies

$$\frac{1}{M}\sum_j q_{u,v}(t_j)\ge\alpha p/2.$$

The normalization argument at the end of the preceding proof gives the claimed stability and a positive lower frame bound. The upper frame bound is finite because each sampled vector has finite norm. □

5 Why the full problem remains open

The source paper's bad L_p discretization spaces for $p<2$ do not produce a counterexample here. After L_2 normalization, their evaluation vectors have size $N^{1-p/4}$, which exceeds $\beta\sqrt{N}$ whenever $p<2$. Sparse coordinate-pair constructions also fail: making their support sparse enough to obstruct every linear-size selection destroys dimension-free continuous phase stability.

Conversely, (SPR) does not by itself visibly imply (10). A fixed two-dimensional pair may receive its lower bound from a rare, high-amplitude component, even while other components stabilize nearby directions. Such exceptional components can be selected deliberately, so this observation is not a counterexample. Bridging N^2 to N appears to require a structural sparsification theorem for the strong complement property, rather than a black-box independent-sampling estimate.

The accompanying script `code/check_scalar_identity.py` checks (1) numerically. It is a sanity check only.

References

[1] D. Freeman and D. Ghoreishi, *Discretizing L_p norms and frame theory*, arXiv:2109.14454; J. Math. Anal. Appl. 519 (2023), 126846.

[2] D. Freeman, T. Oikhberg, B. Pineau, and M. A. Taylor, *Stable phase retrieval in function spaces*, arXiv:2210.05114.

Source context

Page 12 of the locally rendered arXiv:2109.14454 source contains Problem 4.5 and the authors' explanation of its connection to simultaneous L_1/L_2 discretization.

construct a frame of ℓ_2^N which does C-stable phase retrieval. However, there are random constructions where it is possible to choose $C > 0$ such that a frame $(x_j)_{j=1}^m$ of random vectors does C-stable phase retrieval with high probability and the number of vectors m can be chosen on the order of the dimension N [CL14][EM13][KL18][CDFF21]. Each of these results can be thought of as sampling a continuous Parseval frame over a probability space which does stable phase retrieval to obtain a frame which does stable phase retrieval. This naturally leads to the following problem.

Problem 4.5. *Let $C, \beta > 0$. Does there exist constants $D, \kappa > 0$ so that for all $N \in \mathbb{N}$ there exists $M \leq DN$ such that the following holds? Suppose that H is an N-dimensional Hilbert space, (Ω, μ) is a probability space, and $(x_t)_{t\in\Omega}$ is a continuous Parseval frame of H which does C-stable phase retrieval such that $\|\psi_t\| \leq \beta\sqrt{N}$ for all $t \in \Omega$. Then there exists a sequence of sampling points $(t_j)_{j=1}^M \subseteq \Omega$ such that $(\frac{1}{\sqrt{M}} x_{t_j})_{j=1}^M$ is a frame of H which does C-stable phase retrieval.*

This problem seems particularly difficult as Theorem 4.4 which relies on [MSS15] can be thought of as a random sampling result which produces a good frame with low but positive probability. However, all known methods of producing frames which do stable phase retrieval using a number of vectors on the order of the dimension use sub-Gaussian random variables where a random sampling will produce a good frame with high probability. In the following theorem we connect the problem of constructions of frames which do stable phase retrieval to the problem of discretizing the L_1-norm on a subspace of $L_1(\Omega)$. In particular, we prove that in order to sample a continuous Parseval frame to obtain a frame which does stable phase retrieval, it is necessary to simultaneously discretize both the L_1-norm and the L_2-norm on the range of the analysis operator.

Theorem 4.6. *Let $(x_t)_{t\in\Omega}$ be a continuous Parseval frame for an N-dimensional real Hilbert space H over a probability space Ω which does κ-stable phase retrieval and $\|x_t\|_H \leq \beta\sqrt{N}$ for all $t \in \Omega$. Let $T : H \to L_2(\Omega)$ be the analysis operator of $(x_t)_{t\in\Omega}$. Suppose that $(t_j)_{j=1}^M \subseteq \Omega$ is such that $(\frac{1}{\sqrt{M}} x_{t_j})_{j=1}^M$ is a frame of H with upper frame bound B and lower frame bound A which does C-stable phase retrieval. Then both the L_2 norm and the L_1 norm on the range of the analysis operator are discretized in the following way for all $y \in T(H)$,*

(1) $A\|y\|^2_{L_2(\Omega)} \leq \dfrac{1}{M}\displaystyle\sum_{j=1}^M |y(t_j)|^2 \leq B\|y\|^2_{L_2(\Omega)}$,

(2) $A^{1/2}B^{-3/2}C^{-3}(1 + A^{-1}\beta^2)^{-3/2}\|y\|_{L_1(\Omega)} \leq \dfrac{1}{M}\displaystyle\sum_{j=1}^M |y(t_j)| \leq B^{1/2}\kappa^3(1 + \beta^2)^{3/2}\|y\|_{L_1(\Omega)}$.

Before proving Theorem 4.6 we will prove the following lemma.

Lemma 4.7. *Let Ω be a probability space and let X^N be an N-dimensional subspace of $L_2(\Omega)$. Suppose $\kappa, \beta > 0$ are such that $\|x\|_{L_\infty} \leq \beta\sqrt{N}\|x\|_{L_2}$ for all $x \in X^N$ and that*

$$\min(\|f - g\|_{L_2}, \|f + g\|_{L_2}) \leq \kappa\Big\||f| - |g|\Big\|_{L_2} \qquad \text{for all } f, g \in X^N. \tag{4.3}$$

Then, $\|x\|_{L_1} \leq \|x\|_{L_2} \leq \kappa^3(1 + \beta^2)^{3/2}\|x\|_{L_1}$ for all $x \in X^N$.